University at Buffalo
Department of Computer Science and Engineering
CSE 473/573 - Computer Vision and Image Processing

Fall - 2026

Project #2

Due Date: 10/29/26, 11:59PM

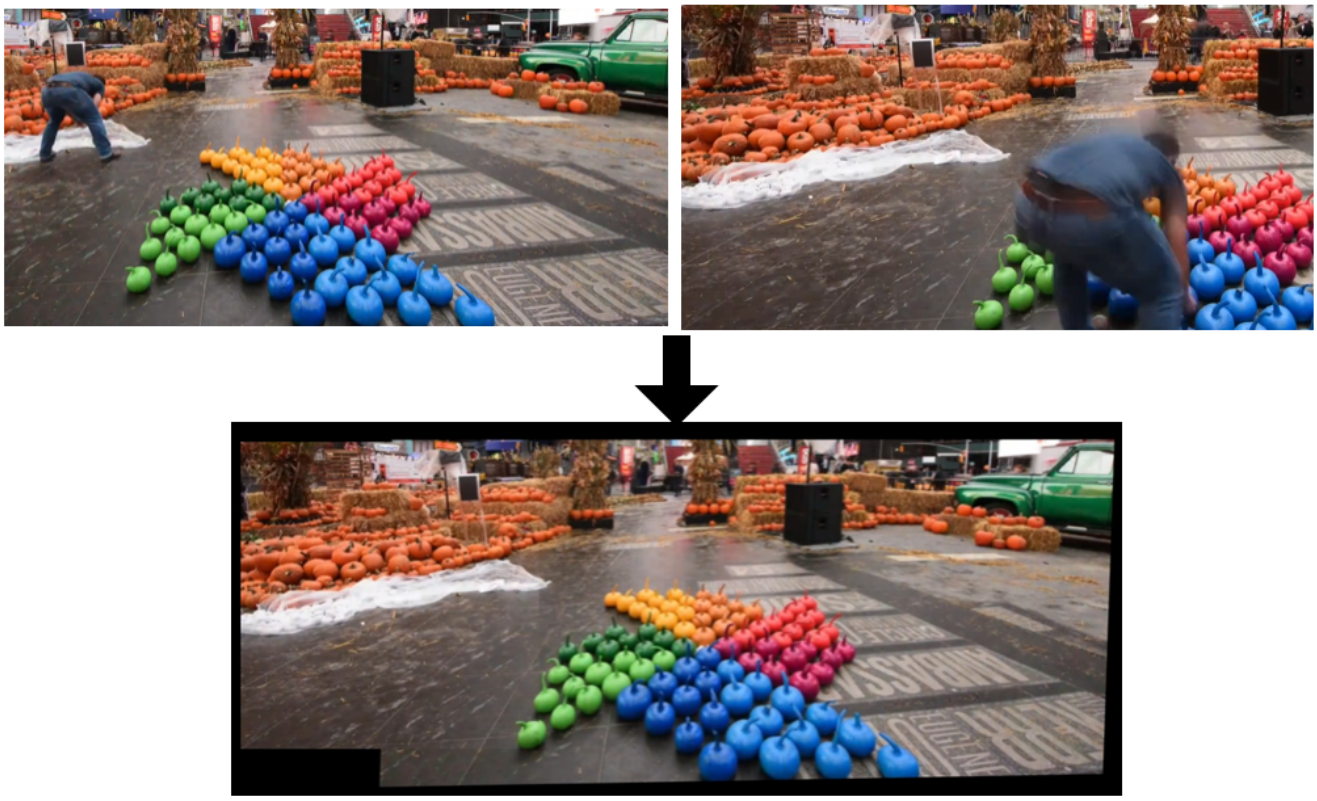


Figure 1: Example of background stitching.

1 Background Stitching (25pts)

The goal of this task is to experiment with image stitching methods. Two images may have the same background but different foreground. For example, a moving person may be moving in the scene. You need to stitch the two images into one image eliminating foreground objects that move

in the scene. You may assume that the region where a foreground object covers the background has similar characteristics to the rest of the background and is visible in two or more images. See the example in Figure 1. The shape of the output image after stitching might be irregular since you cannot crop either of the transformed images. There are no restrictions regarding the method you use to stitch photos into one photo.

Expected Steps:

- Extract a set of key points for each image.
- Extract features from each key point.
- Match features and use matches to determine if there is overlap between given pairs of images.
- Compute the homography between the overlapping pairs as needed. Use RANSAC to optimize your result.
- Transform the images and stitch the two images into one mosaic, eliminating the foreground as described above.
- Save the resulting mosaic according to the instructions specified in the script. (25 points)

2 Image Panorama (25pts)

This task aims to stitch multiple images into one panoramic photo. As shown in Figure 2, the given images might be non-overlapping or multiply-overlapped (overlapping two or more other images). The shape of the output image after stitching might be irregular since you cannot crop either of the transformed images. There are no restrictions regarding the method you use to stitch photos into a panoramic photo. For this project, you can assume the following:

- Your code will need to be able to stitch together four or more images, and you will not know that in advance.
- You can assume that IF an image is to be part of the panorama, it will overlap at least one other image by at least 20%.
- Images that do not overlap with any other image can be ignored.
- Images can overlap with multiple images.
- Although the Figure below shows horizontal panoramas, your five images can be stitched together in any way.
- You are only expected to produce one overall image.
- While some of the most modern techniques may use a spherical projection for better panoramas, you are free to assume that basic 2D planar transformations are sufficient for this project.

Expected Steps:

- Extract features for each image, match features, and determine the spatial overlaps of the images automatically. For instance, among four images, if image1, image3 and image4 overlap with each other, then your overlap array should be $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$. Follow the instructions in the provided script and save the overlapping result as a $N * N$ one-hot array in the JSON file. (8 points)
- Conduct image transformation and stitch into one panoramic photo. Save the photo as the instructions specified in the script. (17 points)

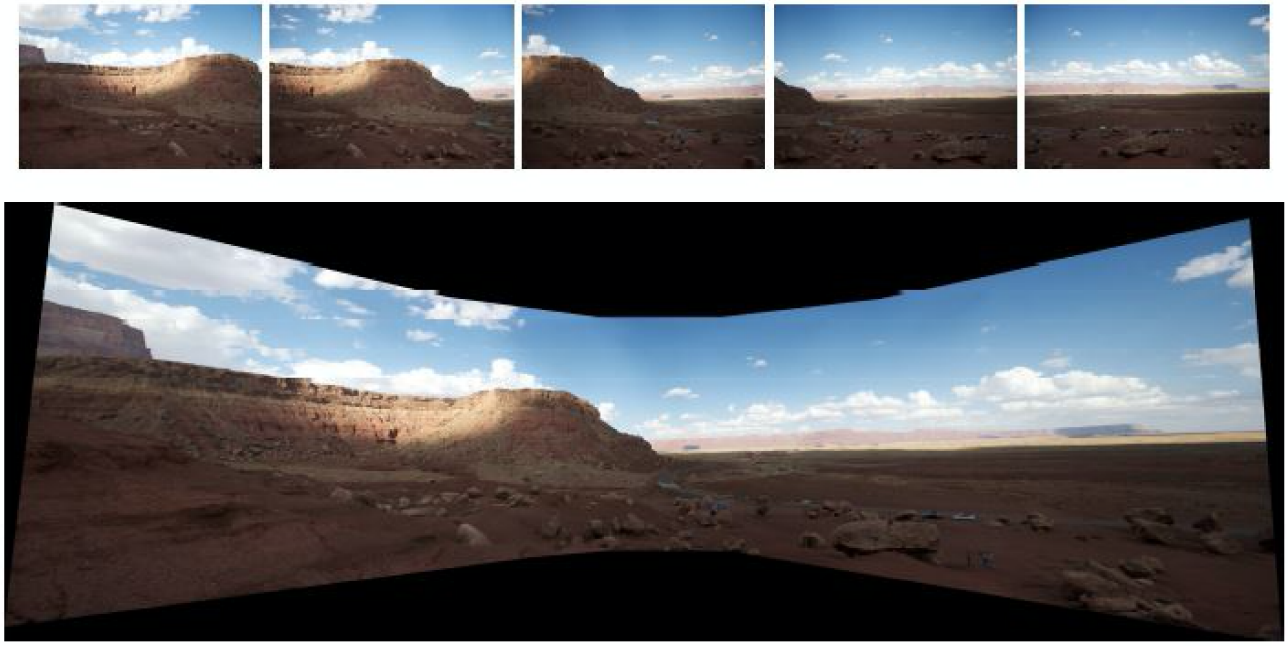


Figure 2: Example of image panoramas.

3 Hidden Test Panorama and Stiching (50pts Hidden)

This is phase 2 of you project 2 where we test 2 hidden tasks same as the Task 1 and Task 2. The images for this task will be hidden and getting as close to it as possible is the goal. This phase tests the generalizing capabilities of your solution.

Scoreboard metrics are same as Project 1 that is Error, Time and Space complexity where lower in all three cases is the best. Dont worry about the ranking in the score board, make sure to reduce all your scores as low as possible and you are good.

Distribution of the points is as follows: 60% Error + 20% Time + 20% Space.

4 Project Guidelines and Submission

- Please Implement all your code in the file **"stitching.py"**. Please do NOT make any changes to any file except **"stitching.py"**.
- Please do NOT read/write any files in your code. The file reading and output part are already given in **"task1.py"** and **"task2.py"**. In your implementation, you should ONLY use the input parameters of the function and give the output in a required data structure. The data structure is given in the code.
- We have provide a library called kornia, please read the environment preparing part for installation. **You are only allowed to use Kornia to complete this project.** Here is document link for kornia: `https://kornia.readthedocs.io/en/stable/`.
- Please do not use cv2, numpy, PIL, imageio for any image related operations. We have provided the required helper functions. If you see any custom methods being useful for your implementation, then please feel free to implement them with PyTorch.
- You can only use the given libraries provided in code. You can not make any new imports.
- Unlimited number of submissions is allowed and only the latest submission will be used for grading.
- To submit your code and result, Please run **"pack submission.sh"** to pack your code and result into a zip file. You can find the command line in "README.md" Note that when packing your submission, the script would run your code before packing. The resulting zip file is the only file you need to submit.
- The packed submission file should be named "submission_YourUBITName.zip", and it should contain 4 files, named "task2.json", "stitching.py", and "images/" folder and "outputs/" folder. If there is something wrong with your code/filename, please go back and check.
- For code raising "RuntimeError", the grade will be ZERO for the project if it can not be corrected. Not following the name rules "UBID.zip" will result in a reduction in grade.
- You have up to 3 late days for all the project/homework submission throughout the semester without points deducted.
- You should NOT copy code from other projects on the internet.
- Identical code will be treated as plagiarism. Please work it out independently. Checks will be done for AI violations. If your submission is evaluated as plagiarism, your grade will be 0 for that task.

5 Preparing for the environment

- **Please note:** In this project, it is highly recommend to use anaconda to manage your environment. Here is a link about miniconda usage: `https://docs.anaconda.com/free/miniconda/index.html`.

5.1 Install kornia

The installation of kornia will require you have anaconda: (It is possible to install it without anaconda, but follow the following step without anaconda may impact your other work in your environment) Also, the face_recognition can be install in same environment.

- Create a new environment

```
$ conda create —name cse473
```

- install python 3.10

```
$ conda install python==3.10
```

- install pytorch 1.11.0 with cudatoolkit==11.3 (You can install with your own way, but the pytorch version should be 1.11.0)

```
$ conda install pytorch==1.11.0 torchvision==0.12.0 cudatoolkit=11.3 -c pytorch
```

- install kornia

```
$ pip install kornia==0.7.0 —no-deps
```

- install matplotlib

```
$ pip install matplotlib
```

We have tested other combination of the library version, if you are getting some warning about cross() from torch and the code located in kornia, it means your pytorch version is wrong.